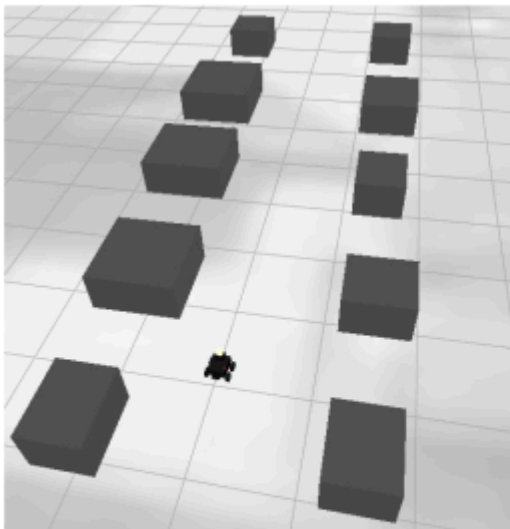
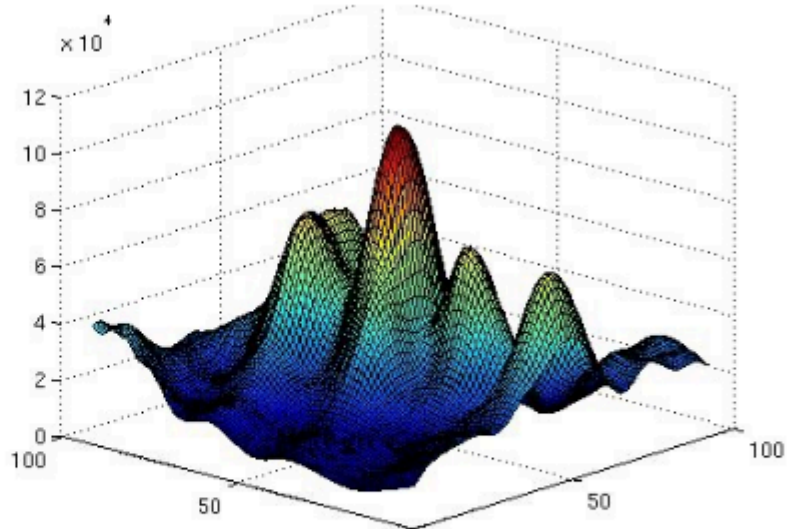


L19: Robust SLAM

- Robust Least Squares
 - Normal: No outliers and association ambiguities
 - But:
 - Outliers occur in real world
 - Wrong Data association:
 - Places that look identical
 - Similar rooms in same building
 - Cluttered scenes
 - GPS multi path (signal reflections)



3D world



belief about the robot's pose

MaxMixtures or Dealing with Multiple Modes

- Mathematical Model
 - Constraints modeling Gaussian noise
 - Instead of computing the sum of Gaussians:
 - $p(z|x) = \eta \exp(-\frac{1}{2}e_{ij}^T \Lambda_{ij} e_{ij})$
 - cannot log inside the sum
 - **Compute the maximum of the Gaussians**

$$p(z|x) = \sum_k w_k \eta_k \exp\left(-\frac{1}{2} e_{ij_k}^T \Lambda_{ij_k} e_{ij_k}\right)$$

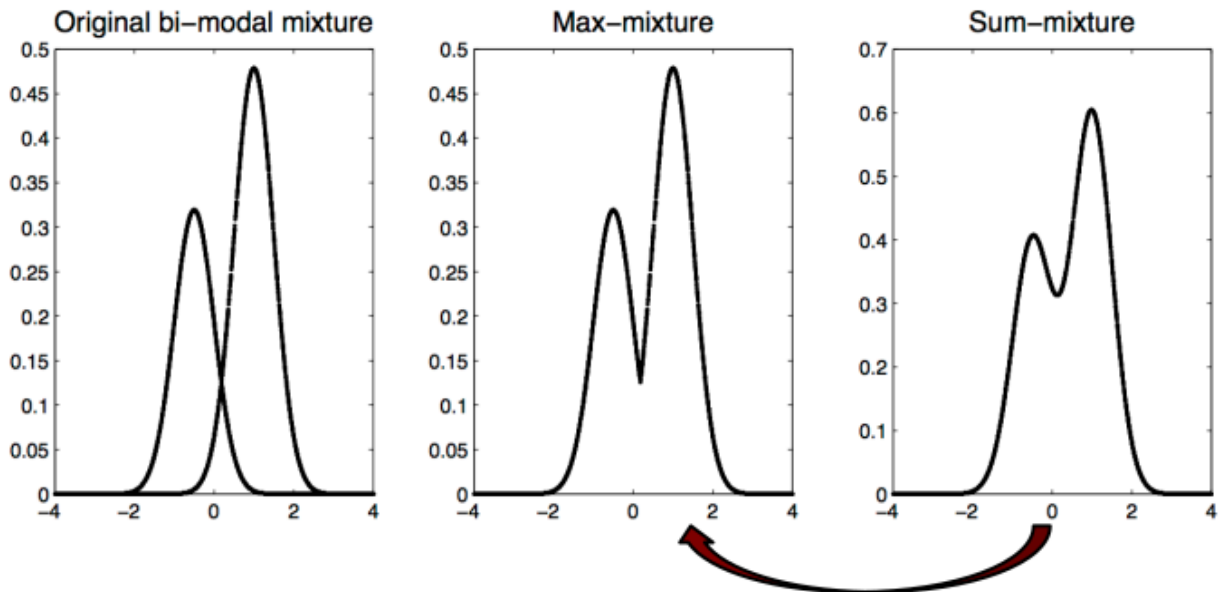
$$\simeq \max_k w_k \eta_k \exp\left(-\frac{1}{2} e_{ij_k}^T \Lambda_{ij_k} e_{ij_k}\right)$$

- log can be moved inside max operator

$$\log p(z|x) \simeq \max\left(-\frac{1}{2} e_{ij_k}^T \Lambda_{ij_k} e_{ij_k} + \log(w_k \eta_k)\right)$$

$$-\log p(z|x) \simeq \max\left(\frac{1}{2} e_{ij_k}^T \Lambda_{ij_k} e_{ij_k} - \log(w_k \eta_k)\right)$$

- Max-Mixture Approximation



approximation error

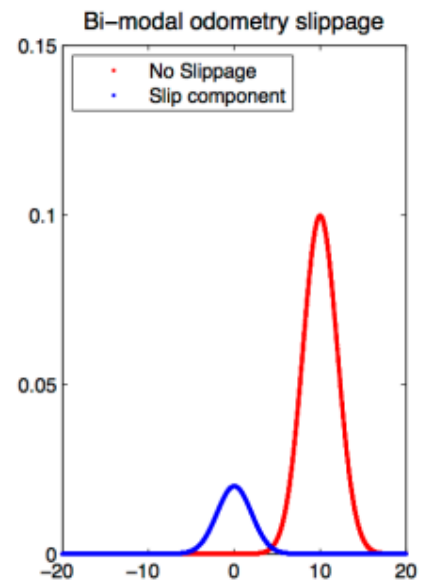
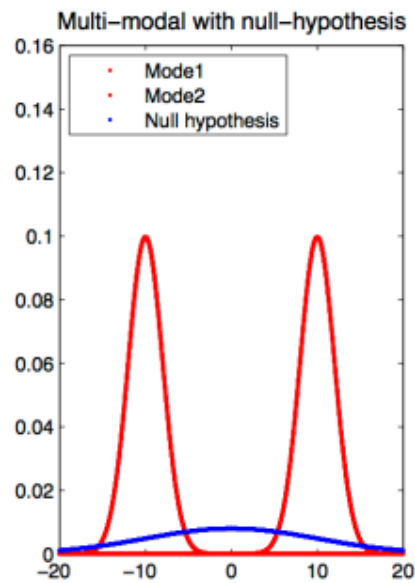
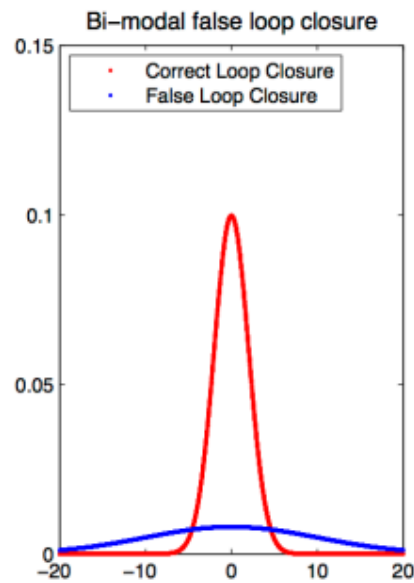
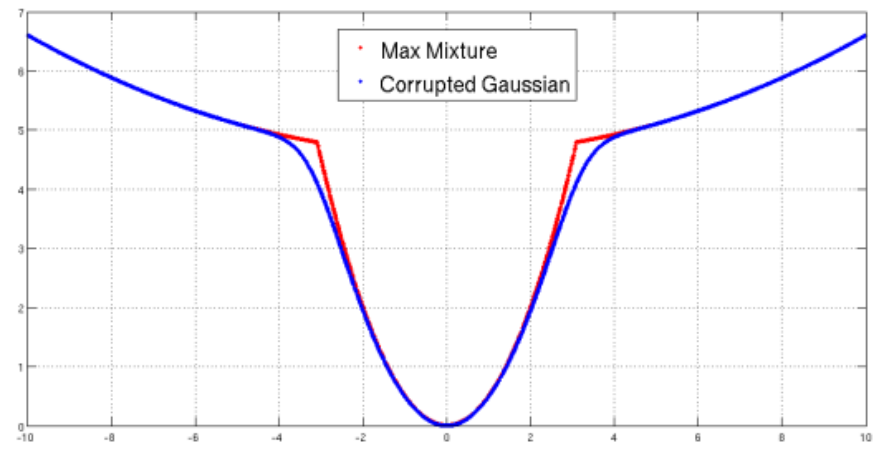
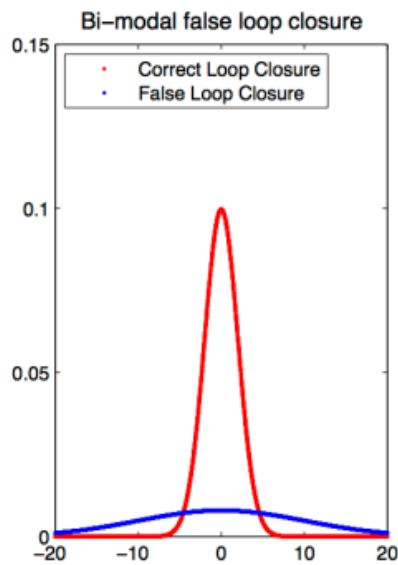
- Integration

- With Max-Mixture formulation:

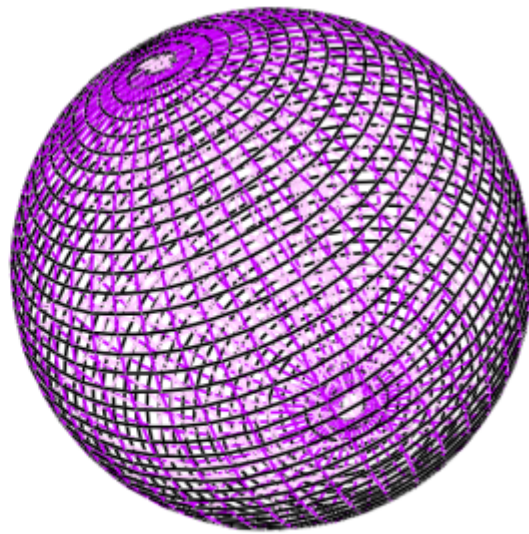
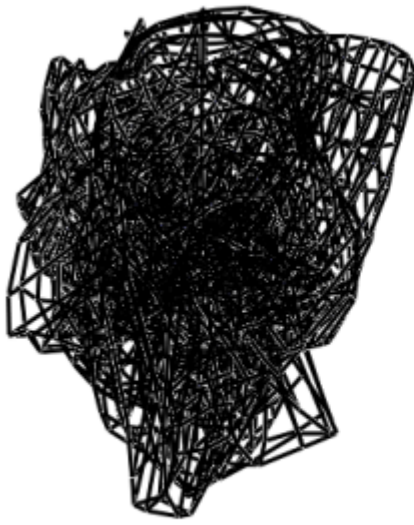
- Log likelihood results in local quadratic forms

- Easy to integrate in optimiser:

1. Evaluate all k components
2. Select component with maximum log likelihood
3. Perform optimisation as before using only the max components (as a single Gaussian)



- MM formulation is useful for multi-model constraints (D.A. ambiguities)
- MM is also handy tool for dealing with outliers
- Outliers:
 - One mode represents the main constraint and
 - 2nd model uses a flat Gaussian for the outlier hypothesis



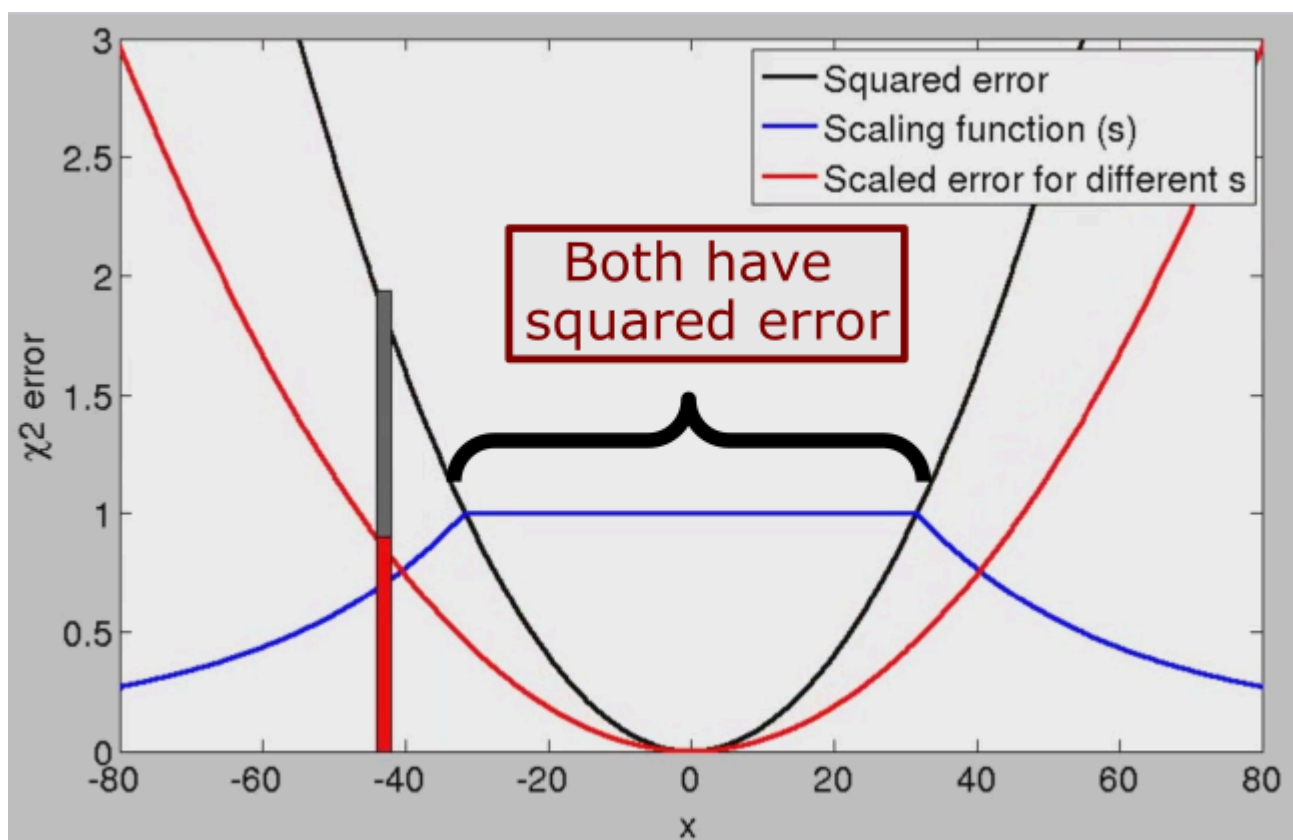
• Gauss-Newton

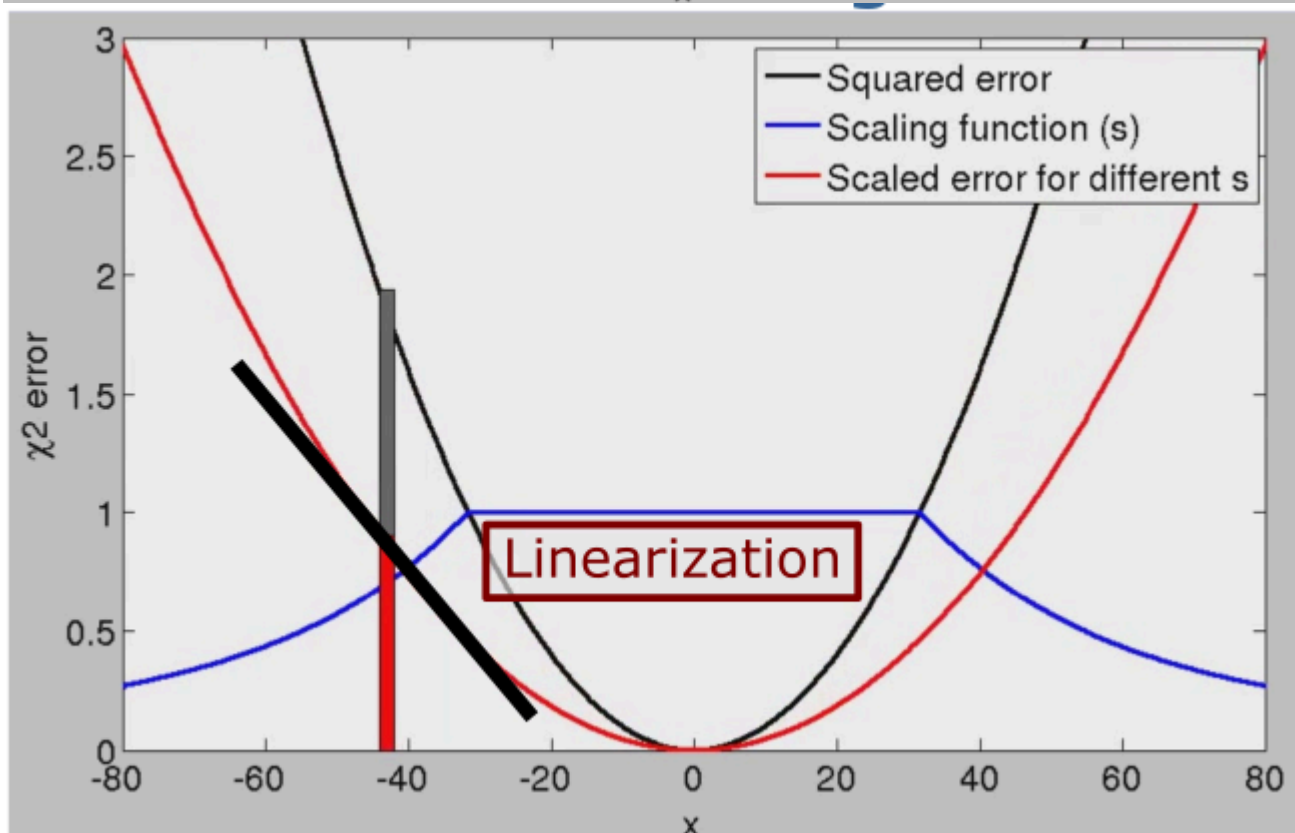
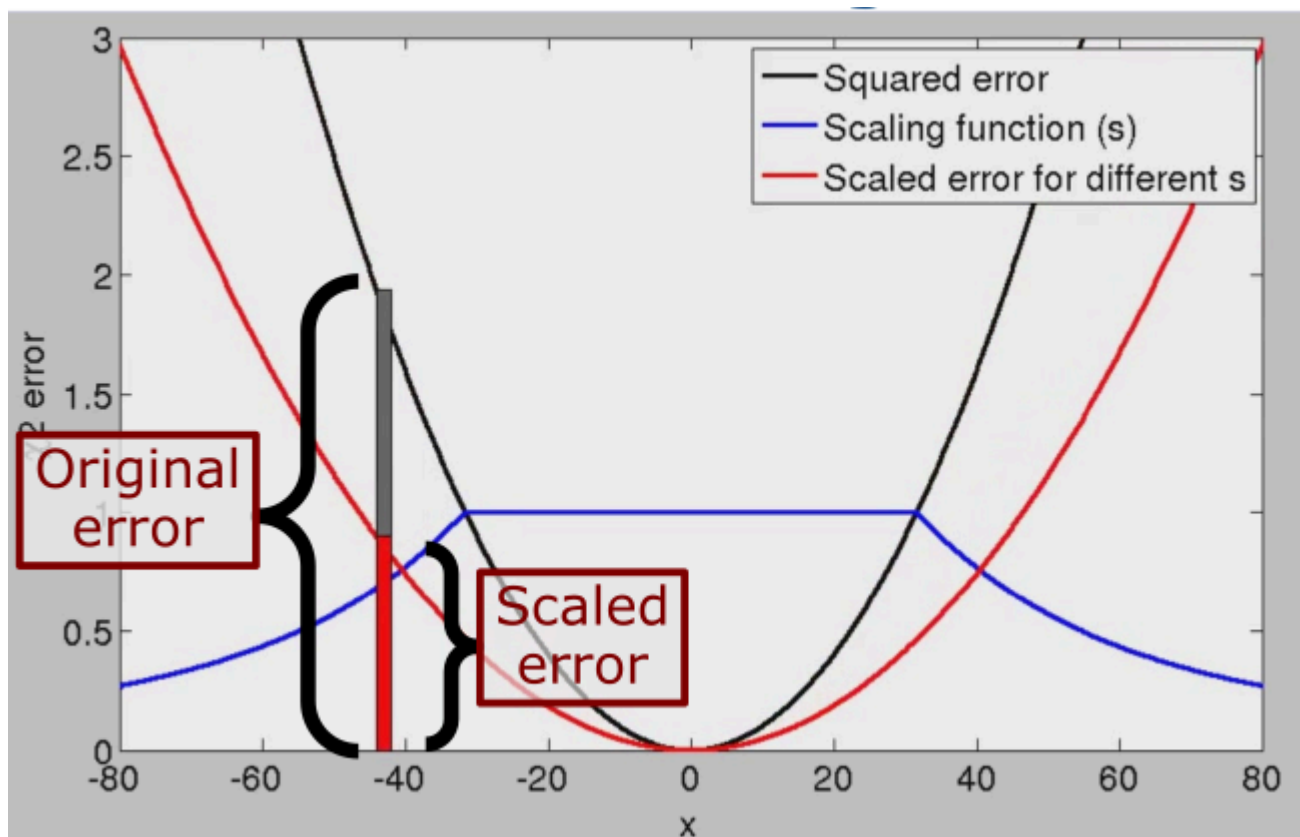
- Here: 100 outliers: the more the worse
- MM: Easy to use, quite effective
- Supports multi-model constraints
 - Select best mode of a sum of Gaussians and use it as if it would be a single Gaussian

MM Gauss-Newton

Dynamic Covariance Scaling

- $x^* = \arg \min_x \sum_{ij} \underbrace{e_{ij}^T(x) \Lambda_{ij} e_{ij}(x)}_{\chi_{ij}^2}$
- $x^* = \arg \min_x \sum_{ij} e_{ij}^T(x) (s_{ij}^2 \Lambda_{ij}) e_{ij}(x)$
 - with $s_{ij} = \min \left(1, \frac{2\Phi}{\Phi + \chi_{ij}^2} \right)$





- Add additional weighting term to the error function
- Weight depends on error value
- Idea: Weight down constraints that are far away from the mean estimate
- Special case of robust least squares estimation
 - Geman-McClure kernel

Least Squares with Robust Kernels

- More probability mass in the tails of the distribution

- Robust M-Estimators:

- $\rho(e)$ function used to define PDF

- $p(e) = \exp(-\rho(e))$

- Minimising the neg.log likelihood

- $x^* = \operatorname{argmin}_x \sum_i \rho(e_i(x))$

- Gaussian: $\rho(e) = e^2$

- Absolute values (L1 norm): $\rho(e) = |e|$

- Huber M-estimator:

- $\rho = \begin{cases} \frac{e^2}{2} & \text{if } |e| < c \\ c(|e| - \frac{c}{2}) & \text{otherwise} \end{cases}$

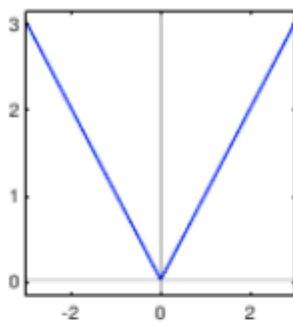
- Mixture of quadratic and linear function

- Quadratic around the solution (noise)

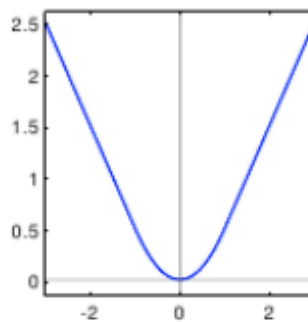
- Linear for outliers (error > threshold)

- Several others

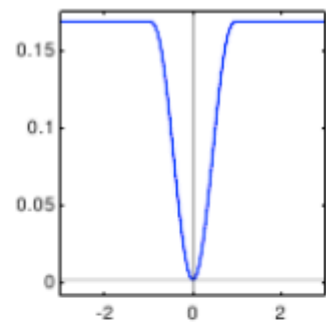
- Tukey, Cauchy, Blake-Zisserman, Corrupted Gaussian



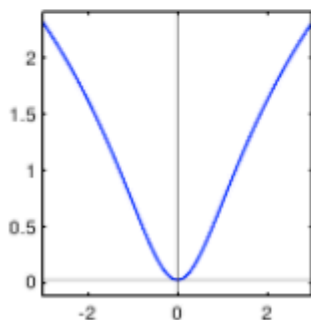
L1 norm



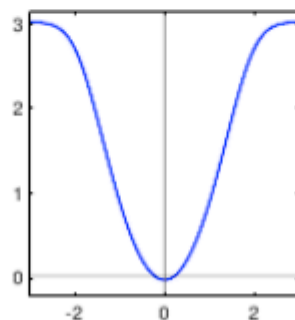
Huber



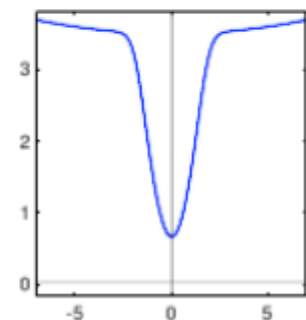
Tukey



Cauchy



Blake-Zisserman



Corrupted G.

Robust Estimation as Weighted Least Squares

- Weighted Least Squares

- $x^* = \arg \min_x \frac{1}{2} \sum_{i=1}^N w_i ||e_i(x)||^2$
- Gradient at optimum goes to zero
 - $\frac{1}{2} \frac{\partial(w_i e_i^2(x))}{\partial x} = w_i e_i(x) \frac{\partial e_i(x)}{\partial x} = 0$

- Robust Estimation

- $x^* = \arg \min_x \sum_{i=1}^N \rho(e_i(x))$
- Gradient at optimum goes to zero
- $\frac{\partial(\rho(e_i(x)))}{\partial x} = \rho'(e_i(x)) \frac{\partial e_i(x)}{\partial x} = 0$

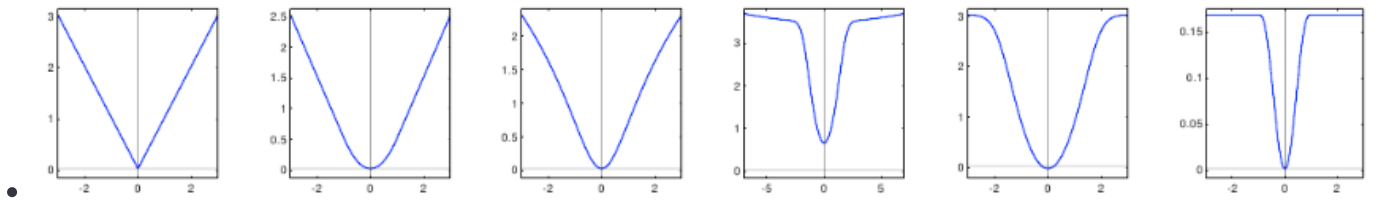
- Robust estimation can use weighted least squares if:

- Set weight using the kernel:
- $w_i = \frac{1}{e_i(x)} \rho'(e_i(x))$

- Kernel will impact Jacobians

- Rest stays the same

- Choice of kernel must align with outlier distribution



Generalised Robust Kernels

- Loss/kernel function selection:

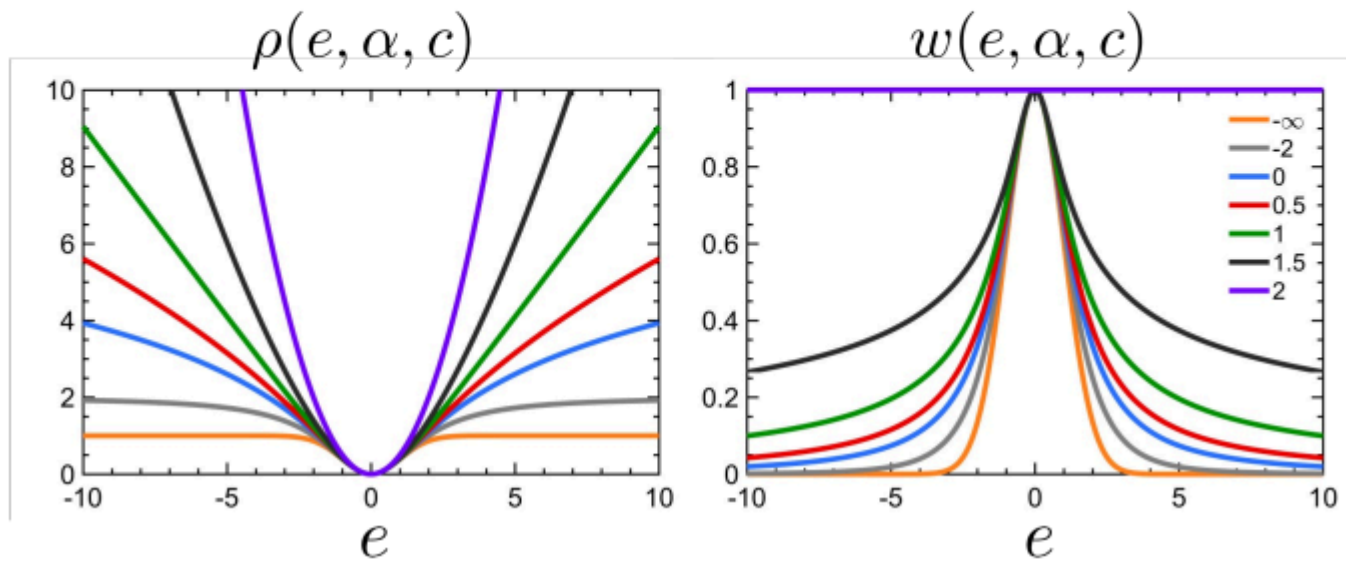
- Depends on type of outliers

- Some approaches combine them:

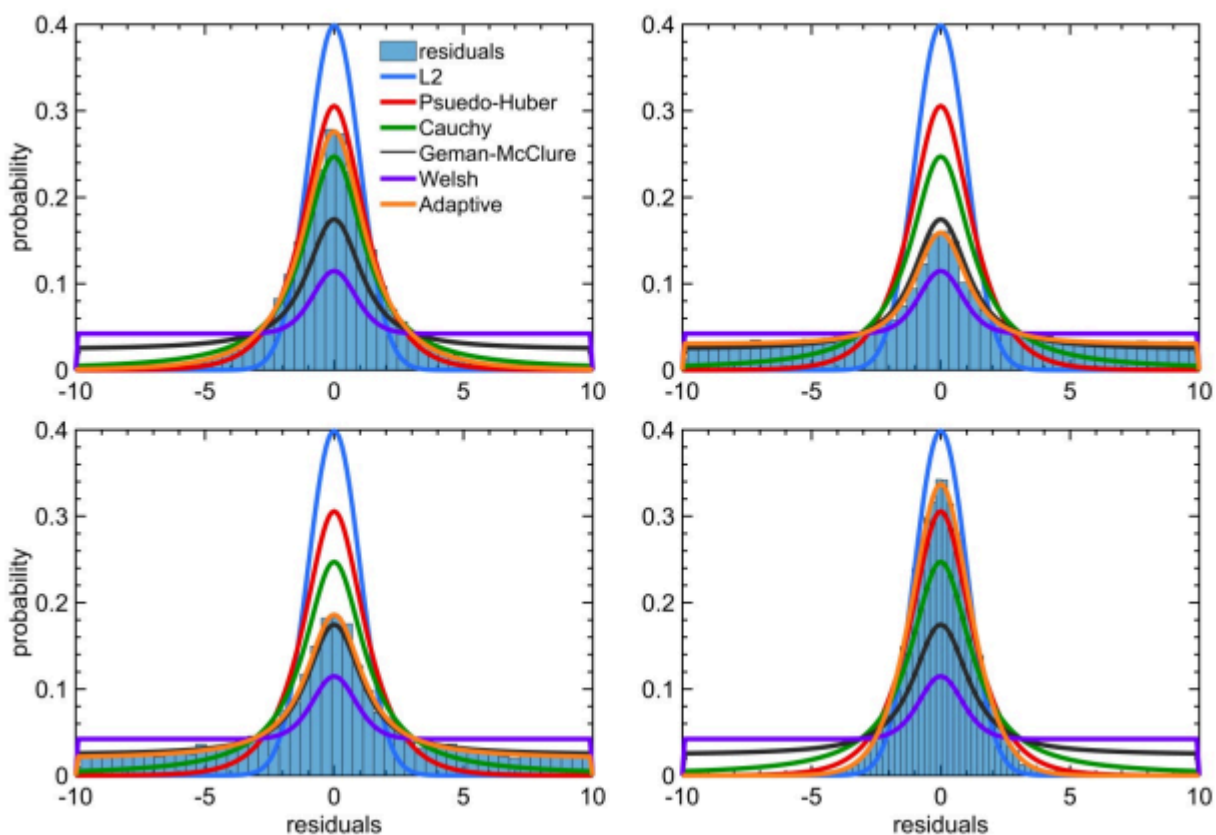
1. Start with strong tails for N_1 iterations
2. N_2 iteration with weaker tails
3. Remove all outliers larger than c
4. Gaussian/Huber for the rest

General Robust Loss Function

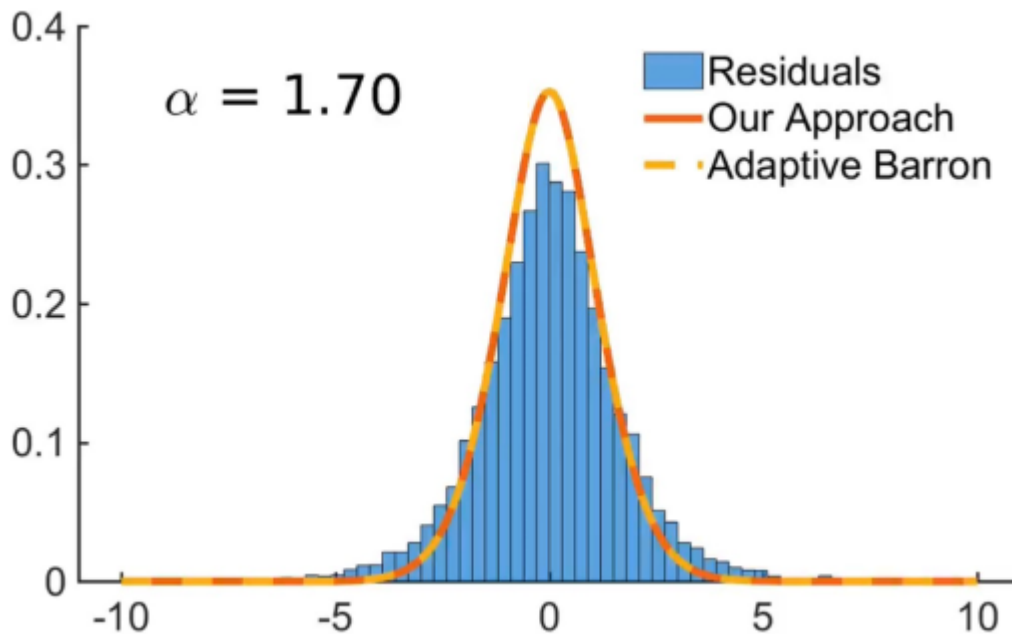
$$\rho(e, \alpha, c) = \frac{|\alpha - 2|}{\alpha} \left(\left(\frac{(e/c)^2}{|\alpha - 2|} + 1 \right)^{\alpha/2} - 1 \right)$$



Adaptive Robust Loss Function



General an Adaptive Robust Loss Function (Barron 2019)



- Jointly optimise over x and α

$$(x^*, \alpha^*) = \arg \min_{(x, \alpha)} \sum_{i=1}^N \rho(e_i(x), \alpha)$$

- Adaptive loss function:

$$\begin{aligned} \rho_a(e, \alpha, c) &= -\log P(e, \alpha, c) \\ &= \rho(e, \alpha, c) + \log cZ(\alpha) \end{aligned}$$

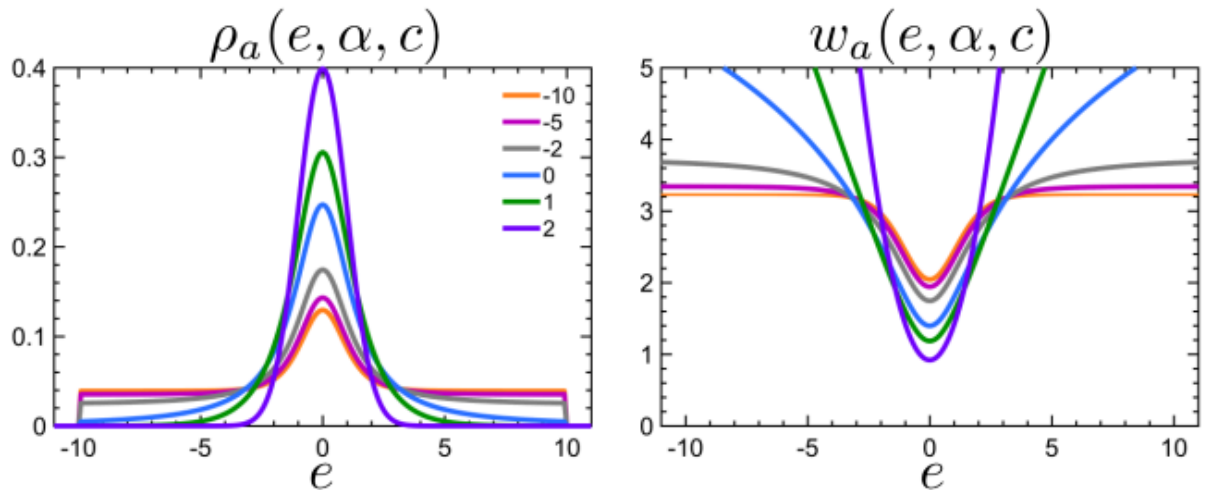
- With Probability Distribution for general loss function

$$\begin{aligned} P(e, \alpha, c) &= \frac{1}{cZ(\alpha)} e^{-\rho(e, \alpha, c)} \\ Z(\alpha) &= \int_{-\tau}^{\tau} e^{-\rho(e, \alpha, 1)} de \end{aligned}$$

- Approaches infinity for negative α , if integral from $-\infty$ to ∞
 - Limit range of outliers to maintain a pdf
 - $Z(\alpha) = \int_{-\tau}^{\tau} e^{-\rho(e, \alpha, 1)} de$

- Adaptive loss function defined as

$$\begin{aligned}\rho_a(e, \alpha, c) &= -\log P(e, \alpha, c) \\ &= \rho(e, \alpha, c) + \log cZ(\alpha)\end{aligned}$$



- With this: solve joint optimisation problem
 - in weighted least squares use:
 - $\rho_\alpha(e, \alpha, c)$ as robust kernel
- Problems in practise:
 - New Jacobians need to be computed
 - α can dominate the parameter estimation for complex problems
 - Sensitive to initial guess

Alternating Optimisation

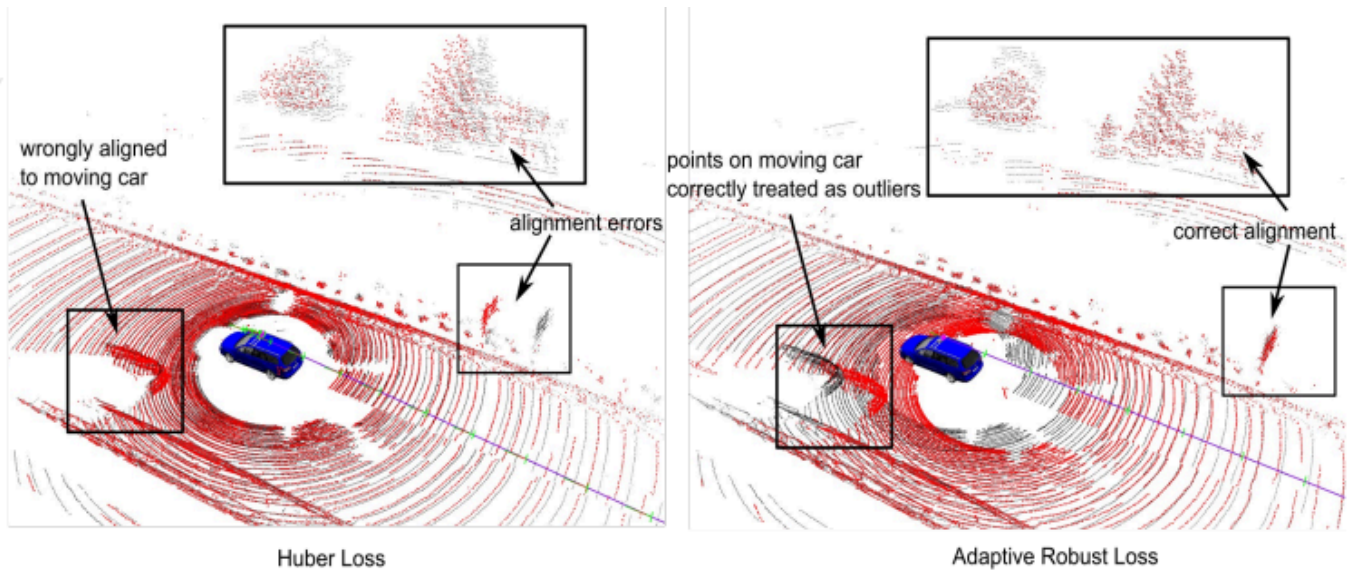
- Solve via alternating both steps
- 1D line search problem
 - $\alpha^t = \arg \max_{\alpha} \sum_{i=1}^N \log P(e_i(x^{t-1}), \alpha^{t-1}, c)$

- Minimise as weighted least squares

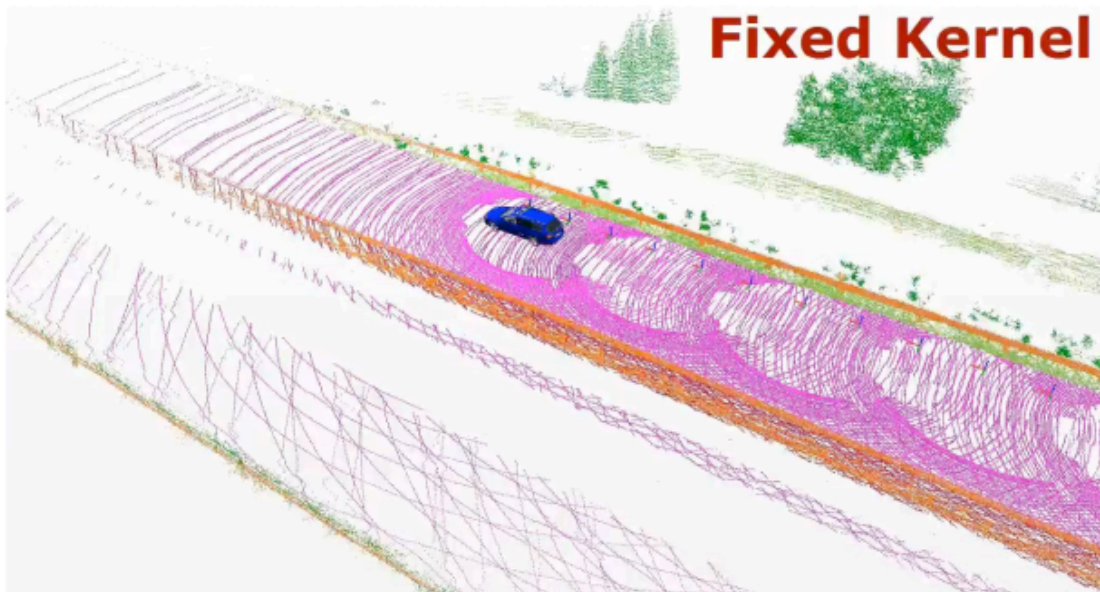
- $x^t = \arg \min_x \sum_{i=1}^N \rho_\alpha(e_i(x), \alpha^t, c)$

- Kernel and unknowns estimated in iterative fashion
- Alternating estimation provides better results than joint optimisation
- Kernel function adapts to current situation by adapting α
- No need to change Jacobians

ICP Example



Outlier rejection in presence of dynamic objects



Dealing with Outliers - Summary

- Max-Mixture support multi-model constraints
 - Approximates sum of Gaussians using max operator
- Dynamic Covariance Scaling: For fixed kernel in SLAM
 - DCS is a form of Geman-McClure
- Robust kernels are key
- Choice of robust loss function depends on given problem
 - Popular: Huber, L1, DCS/Geman McClure
- Better: Don't commit on one kernel
- Adaptive robust kernel is the most flexible way

- Best results: Adaptive kernels in alternating estimation